

9. Tecniche di valutazione I: Observational Methods

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Evaluation

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- requires an artifact: simulation, prototype, full implementation
- tests usability and functionality of system
- occurs in laboratory, field and/or in collaboration with users
- evaluates both design and implementation
- should be considered at all stages in the design life cycle

Goals of Evaluation

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- to assess extent of system functionality
- to assess effect of interface on user
- to identify specific problems

Valutazione mediante la partecipazione degli utenti

- partecipanti: rappresentativi degli utenti
- esperienza simile tra loro (omogenei)
- conoscenza di quel tipo di sistemi e del dominio
- numerosità dei partecipanti: J. Nielsen suggerisce almeno 3 / 5 utenti

Scenario e task

- lo scenario descrive una situazione potenzialmente reale nella quale si immedesimano i partecipanti alla valutazione
- permette di individuare: utenti, azioni, strumenti da usare e contesti
- i task sono i singoli compiti che si richiede di svolgere ai partecipanti
- è possibile anche svolgere valutazioni di uso libero del sistema, senza task (ma non nel progetto!)

Stili di valutazione

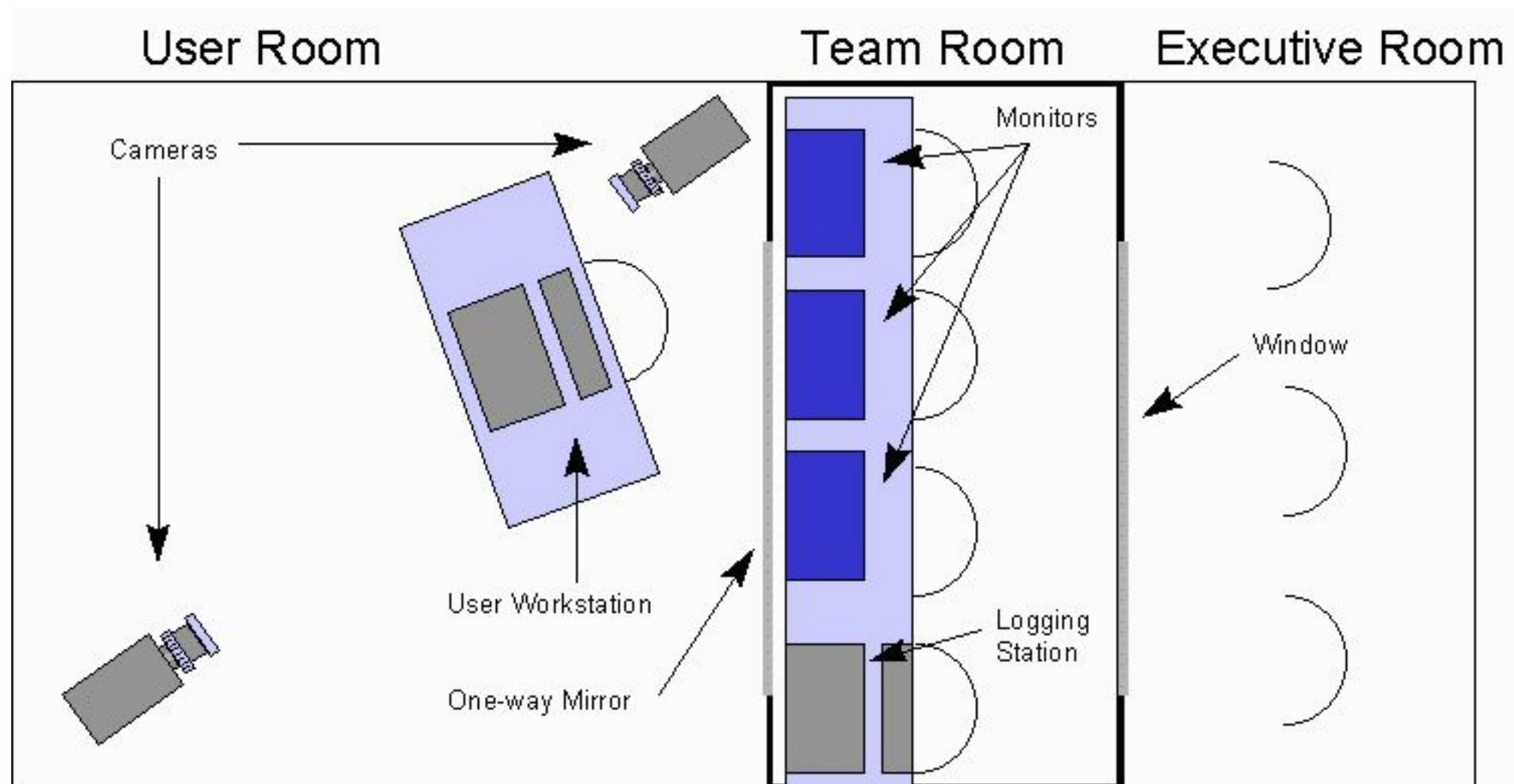
- in laboratorio (lab)
- sul campo (field)

Laboratory studies

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- + specialist equipment available
- + uninterrupted environment
- lack of context
- difficult to observe several users cooperating
- Appropriate if system location is dangerous or impractical

Laboratorio

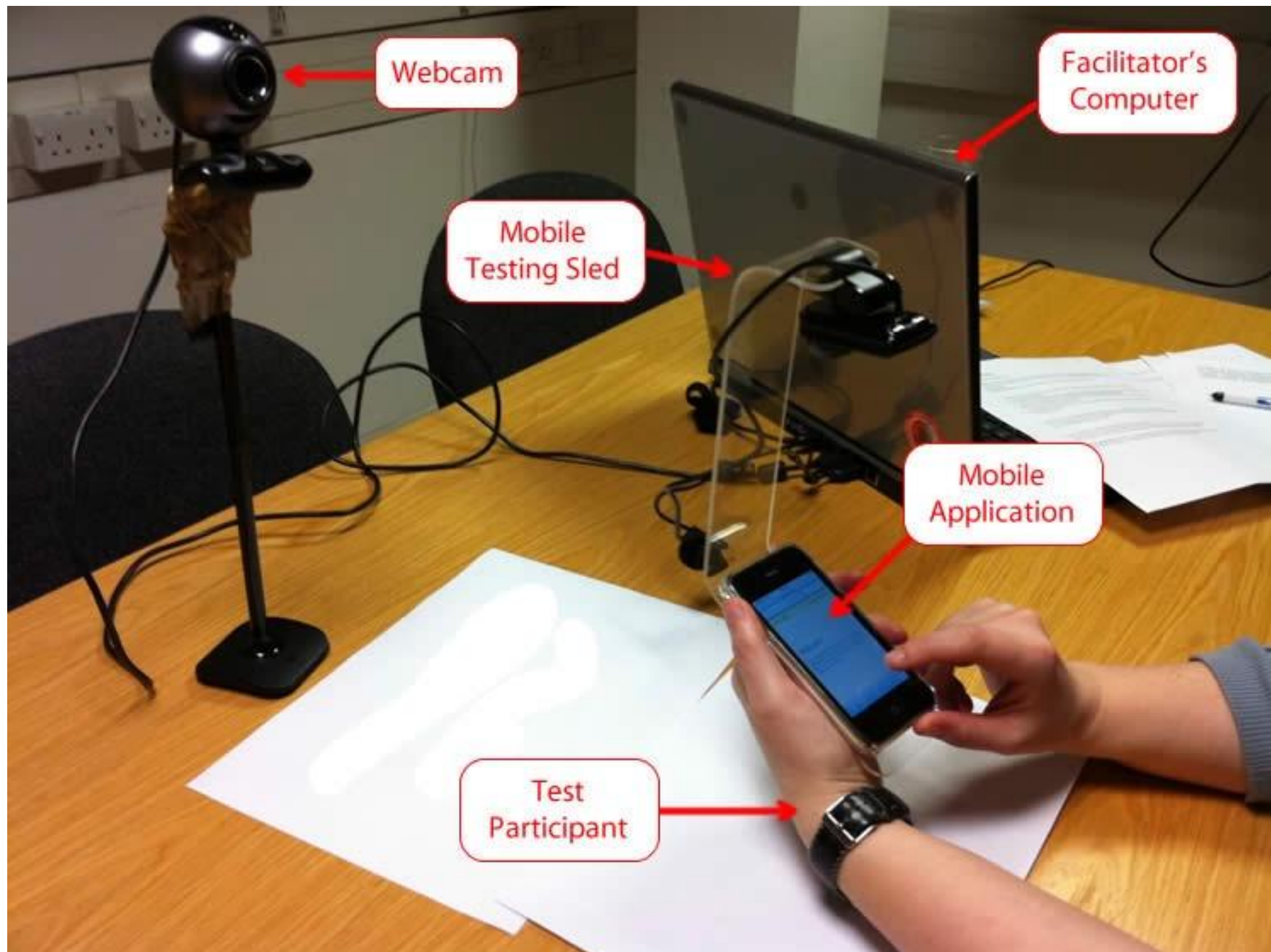




Laboratorio



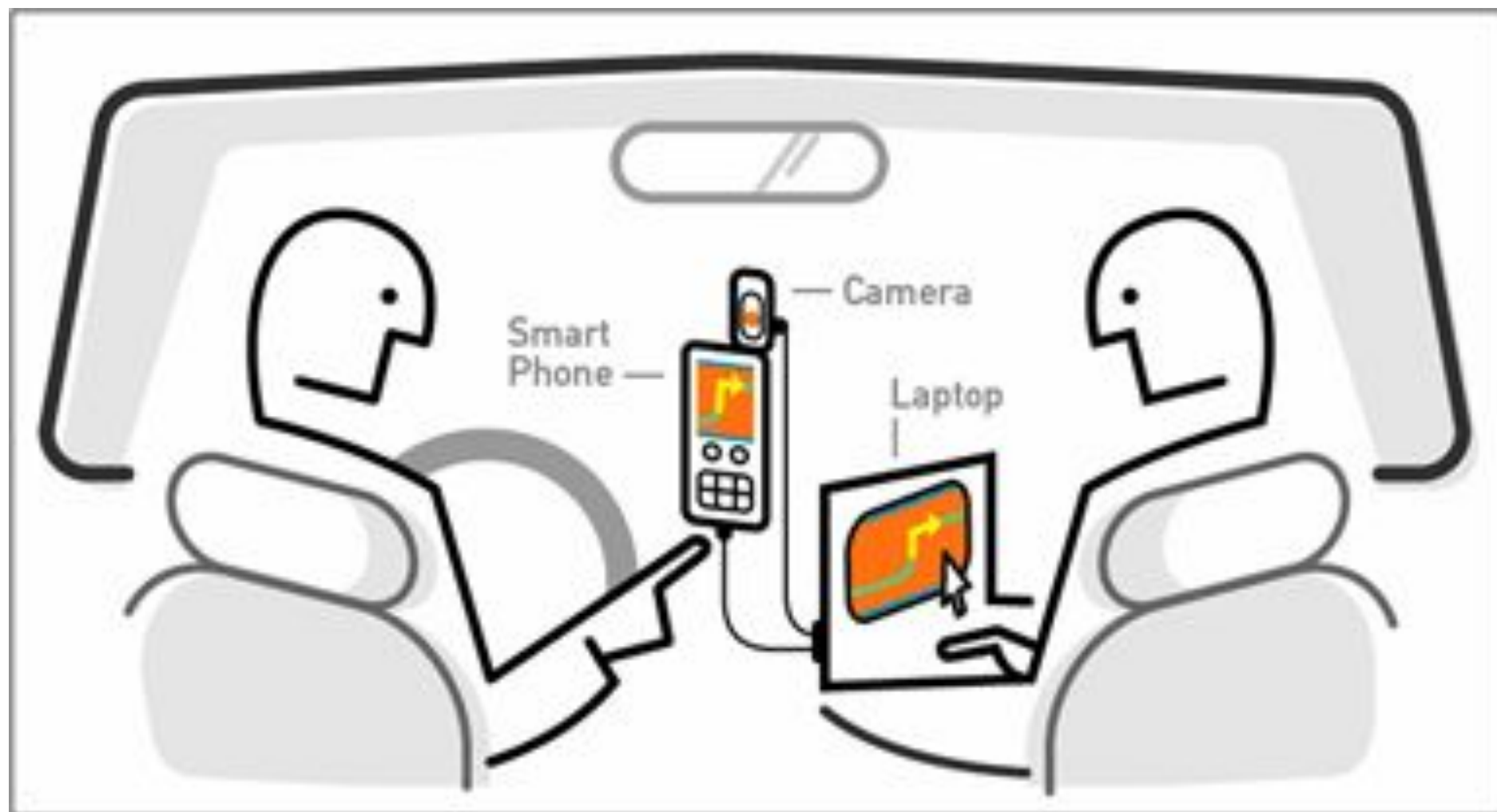
Laboratorio



Field Studies

- + natural environment
- + context retained (though observation may alter it)
- + longitudinal studies possible
- distractions
- noise
- Appropriate where context is crucial for longitudinal studies

Field



Field



Observational Methods

- Think Aloud
- Cooperative evaluation
- Protocol analysis
- Post-task walkthroughs

Think Aloud

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- user observed performing task
- user asked to describe what he is doing and why, what he thinks is happening etc.

Think Aloud: vantaggi e svantaggi

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- + simplicity - requires little expertise
- + can provide useful insight
- + can show how system is actually used
- subjective
- selective
- act of describing may alter task performance

Cooperative evaluation

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- variation on think aloud
- user collaborates in evaluation
- both user and evaluator can ask each other questions throughout
- Additional advantages:
 - + less constrained and easier to use
 - + user is encouraged to criticize system
 - + clarification possible

Protocol analysis

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- protocol = recording of evaluation session
 - paper and pencil – cheap, limited to writing speed
 - audio – good for think aloud, difficult to match with other protocols
 - video – accurate and realistic, needs special equipment, obtrusive
 - computer logging – automatic and unobtrusive, large amounts of data difficult to analyze <https://www.youtube.com/watch?v=ConsSIlf6n4>
 - user notebooks – coarse and subjective, useful insights, good for longitudinal studies
- Mixed use in practice
- Audio/video transcription difficult and requires skill <https://archive.mpi.nl/tla/elan>
- Some automatic support tools available

Post-task walkthroughs

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- transcript played back to participant for comment
 - immediate → fresh in mind
 - delayed → evaluator has time to identify questions
- useful to identify reasons for actions and alternatives considered
- necessary in cases where think aloud is not possible

Questi metodi sono soggettivi

- dipendono dalle conoscenze e capacità del valutatore
- il valutatore deve riconoscere i problemi e capire cosa fa l'utente
- evaluator bias: lo si può attenuare mediante l'uso di più valutatori
- anche la risposta dell'utente è soggettiva

Misure qualitative

- misure prodotte da questi metodi: non numeriche
- possono rivelare dettagli non determinabili da misure numeriche
- generalmente i metodi soggettivi portano misure qualitative

Ethical issues

- test anonimi (id anonimi, informazioni aggregate)
- ricompense per i partecipanti

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